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ALGERIAN MINISTRIES WEB QUALITY INDEX 2026

A Standards-Based Evaluation of Security, Performance, Accessibility, Reliability, and Overall Web Quality

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1 Introduction and Objectives

The Algerian Ministries Web Quality Index 2026 is a reproducible national benchmark for evaluating official Algerian ministry websites using a transparent technical quality score from 0 to 100. The index is intended to identify strong public-sector web practices, weak or risky implementations, common modernization needs, and actionable recommendations for improvement.

The index measures externally observable web quality. It does not attempt to certify internal security, internal governance, software architecture, legal compliance, or operational maturity that cannot be verified from public evidence.

Project Objectives

The project will produce:

- a national ranking of Algerian ministry websites;
- one technical profile for each ministry;
- comparative statistics across ministries;
- a public deliverables package containing a formal PDF report, interactive website/dashboard, and machine-readable CSV/JSON dataset;
- prioritized recommendations for modernization.

Methodological Position

The score is a project-defined external technical benchmark, not a government certification, legal opinion, penetration test, or complete security audit. Every scored result must be traceable to stored evidence. Criteria that cannot be measured reliably are marked N/M (Not Measured). Automated checks, manual review and AI-assisted analysis may support evidence classification, consistency checks and drafting, but final scores and public claims require traceable evidence and independent human approval. Performance results use repeated measurements, sensitive testing remains passive and non-destructive, and rankings remain provisional until required monitoring, stability and sensitivity checks are complete.

2 Background and Related Work

The project builds on existing e-government and web-quality evaluation work, but its exact scope is distinct. Previous work generally focuses on a narrower slice of the problem, such as usability, accessibility, or security headers. This project combines security, performance, accessibility, reliability, usability, content/service quality, maintenance, web standards, and trust signals into a single documented technical index.

Algeria-Specific Evidence

Public reporting summarized by ANPT on the GAAN Digital Observatory describes a 2021 evaluation of 48 Algerian institutional websites using a 100-point score, with widespread weaknesses and limited HTTPS adoption. Mohamed Benaida's 2023 Algeria–UK usability study examined 10 Algerian e-government services and reported layout, text-presentation and multimedia weaknesses. UN DESA's 2024 E-Government Development Index gives Algeria an EGDI value of 0.5956 and rank 116.

This Algeria-specific evidence is useful, but it does not provide a current ministry-by-ministry benchmark combining the seven categories defined in this methodology. The present project addresses that gap through a common corpus, scoring rubric, evidence model and reproducible measurement protocol.

International Standards and Benchmarking Context

The methodology uses recognized standards, reference models and operational guidance rather than project-only assumptions. WCAG 2.2 and WCAG-EM 2.0 guide accessibility evaluation and representative sampling. ISO/IEC 25010:2023 provides a product-quality reference model. Core Web Vitals and web-performance guidance support performance measurement, while OWASP and browser-security guidance support externally observable hardening checks. UN DESA and public digital-service frameworks provide benchmarking context for government online services. Together, these references support multiple evidence types, repeated measurement and manual review where automated tools cannot determine the complete user outcome.

3 Scope and Evaluation Corpus

This study evaluates public-facing ministry web presence, not the ministries themselves or their internal systems.

Included

- Official public websites of Algerian ministries, after domain verification.
- Public pages reachable without authentication.
- Passive checks of HTTP, TLS, DNS, headers, browser behavior, performance, accessibility, availability and content quality.
- Representative pages, multilingual versions and citizen journeys for each ministry where available.
- Publicly visible services, forms, documents, contact paths, and search/discovery features, without submitting sensitive or destructive actions.

Excluded

- Active intrusion, exploitation, brute force, fuzzing, denial-of-service, password attacks, or authentication bypass attempts.
- Internal infrastructure, non-public administrative systems, and authenticated user areas.
- Legal certification or formal data-protection compliance opinions.
- Judgement of political, editorial, or policy quality beyond website transparency and service usability.

4 Evaluation Corpus Selection

This phase defines how the project will choose the official ministry websites and the representative pages that will be evaluated later. The ministry list is built from official government sources, frozen on a published cut-off date, and versioned so later portfolio or domain changes can be tracked.

Each candidate domain is confirmed using official links, ministry branding, contact information, public references, and technical identity evidence where available. A `.dz` or `.gov.dz` suffix alone is not treated as proof of ownership, and unresolved domains remain outside the ranking until confirmed. If several official sites exist, the main institutional website is used for the national ranking, while related service portals are inventoried separately.

The index does not evaluate homepages only. For every confirmed ministry website, evaluators test a comparable sample where available:

1. homepage;
2. important public-service page;
3. news or publication page;
4. contact page or form page;
5. deep informational page;
6. one reproducibly selected random page from the eligible crawl corpus.

When language versions or service journeys use different templates, representative pages from those implementations are added. Missing page types are marked N/M or replaced with the closest documented substitute. Redirects, exclusions, selected pages, and sampling decisions are stored as evidence so the later evaluation remains reproducible.

5 Measurement Protocol

5.1 Test Conditions and Responsible Collection

Measurements will be collected under documented and repeatable conditions. Each automated run records the date, time zone, location, tested URL, final canonical URL, device profile, network profile, browser and tool versions, cache state, consent/banner state, and number of repetitions. Crawlers must respect `robots.txt`, use conservative concurrency, avoid authenticated or destructive actions, and stop on instability or rate limiting.

Performance tests use repeated mobile runs per sampled page and report the median plus spread. Reliability monitoring uses predeclared endpoints and intervals over the planned monitoring period; planned maintenance and observation gaps are reported separately. Until the monitoring period is complete, affected rankings remain provisional.

5.2 Measurement Tools and Evidence

The implementation may use recognized tools such as MDN HTTP Observatory, SSL Labs, Internet.nl, Lighthouse, PageSpeed Insights, Chrome DevTools, axe, WAVE, DNS tools, passive browser inspection, custom scripts, and a conservative uptime monitor. The final protocol records the exact tool, browser, ruleset, and version used for every run.

Automated output is treated as evidence, not as a final judgement by itself. Manual review remains necessary for accessibility, mobile usability, forms, citizen journeys, content quality, service links, and cases where tools disagree. Raw outputs, screenshots, headers, TLS results, accessibility findings, performance reports, uptime logs, and evaluator notes are stored with the evidence record.

6 Scoring Model and Ranking Rules

Category	Weight
External Security Posture and Technical Privacy	20
Performance	15
Accessibility	15
Reliability and Availability	10
Mobile Usability and Citizen Journeys	15
Content, Public Services and Trust	15
Technical Quality and Findability	10
Total	**100**

6.1 Aggregation, Missing Data and Ranking Stability

Each rubric item is scored first at page level, then aggregated to a ministry value using the predeclared page set. Site-wide controls such as HTTPS enforcement are assessed once, while page-dependent controls use the median or documented worst-important-page rule specified in the scoring workbook. The same defect is deducted once unless it causes two independently defined user outcomes.

The public result includes the raw score, category scores, measurement coverage, status, and observation date. Minimum coverage thresholds must be declared before scoring begins. Scores are not renormalized upward when items are N/M, and ministries with insufficient coverage are not ranked.

Exact ties share a rank, and differences smaller than two points are presented as the same performance band unless repeated-run and sensitivity analysis show the ordering is stable. If a major category is N/M, the ministry is labelled Not Rankable - Retest Required or Provisional, depending on the cause. The ranking dataset lists every unmeasured criterion.

7 Evaluation Rubric

7.1 External Security Posture and Technical Privacy - 20 Points

Check	Points	Evidence
HTTPS enforced	2	HTTP redirect chain, browser URL
Valid TLS certificate	2	Certificate validity, issuer, hostname match
Modern TLS configuration	2	SSL Labs or equivalent TLS report
HSTS present and meaningful	2	Strict-Transport-Security header
Content-Security-Policy present and meaningful	2	CSP header and policy review
Other important security headers	2	Header capture and Observatory-style checks
Secure cookie configuration	2	Set-Cookie flags on observed cookies

Check	Points	Evidence
No mixed HTTP/HTTPS content	1	Browser and crawler evidence
No unnecessary software/version exposure	1	Headers, HTML, Wappalyzer notes
No obvious sensitive configuration exposure	2	Passive crawl only
Third-party tracking and technical privacy restraint	2	Requests, cookies and tracker inventory

Scoring note: a missing CSP is not automatically a severe vulnerability, but it is a meaningful missing hardening control. Sensitive findings should be summarized without exposing direct exploitation paths.

7.2 Performance - 15 Points

Check	Points	Evidence
Largest Contentful Paint	4	Median lab value; 75th percentile field value where available
Responsiveness	3	Field INP where available; median lab TBT as a labelled proxy only
Cumulative Layout Shift	3	Median lab value; 75th percentile field value where available
Server response and first content	2	TTFB and FCP under declared test conditions
Resource and page-weight efficiency	2	Transfer size, requests, image/script diagnostics
Performance consistency	1	Dispersion across repeated runs and sampled pages

Performance must record mobile and desktop contexts separately. The main public score prioritizes mobile because many citizens access services from phones. The Lighthouse aggregate performance score remains useful diagnostic data but is not scored, preventing LCP, CLS and related metrics from being counted twice. Field data and lab data are never silently merged; missing CrUX data does not itself lose points.

7.3 Accessibility - 15 Points

Check	Points	Evidence
Verified WCAG 2.2 issue set	3	Automated findings manually verified
Color contrast	2	Tool results plus manual spot check

Check	Points	Evidence
Image alternative text	1	Automated and manual review
Form labels, instructions and errors	2	Forms inspected manually and by axe
Keyboard navigation	2	Manual keyboard path notes
Semantic HTML and headings	1	DOM/headings outline
Focus indicators	1	Manual interaction notes
Correct page language and ARIA usage	1	HTML lang, ARIA warnings
Zoom, reflow and orientation	1	Manual responsive accessibility check
Complete-process accessibility check	1	Selected citizen journey notes

The project targets WCAG 2.2 Level AA as a reference, but a full conformance claim requires a complete WCAG-EM 2.0 evaluation by qualified reviewers. Automated tool scores are retained as diagnostics only and never receive independent points or imply WCAG compliance.

7.4 Reliability and Availability - 10 Points

Check	Points	Evidence
Uptime percentage	4	Complete 28-day monitor log
HTTP/server failure rate	2	Status and timeout history
Response-time stability	2	Median, p90 and p95 response times
DNS, TLS and redirect continuity	1	Failure events across the connection chain
Critical endpoint consistency	1	Home and selected service endpoint history

Before the monitoring period is complete, rankings are labelled provisional. A site that remains unreachable after retries from at least two networks and a later time window is marked Not Rankable - Retest Required, not assigned zero.

7.5 Mobile Usability and Citizen Journeys - 15 Points

Check	Points	Evidence
Responsive design and reflow	2	Desktop/mobile screenshots
Mobile readability and touch targets	2	Manual viewport review
Navigation and information architecture	2	Task walkthrough notes

Check	Points	Evidence
Forms usable on mobile	2	Manual form inspection
Citizen-journey task success	4	Standard task protocol
Interface consistency	1	Sample-page comparison
Errors, recovery and contextual help	2	Journey and form notes

7.6 Content, Public Services and Trust - 15 Points

Check	Points	Evidence
Official identity and domain ownership clear	1	Government sources and site evidence
Ministry mission and organization available	1	About/organization pages
Public services clearly listed	2	Service directory evidence
Procedures and eligibility explained	2	Procedure pages or documents
Required documents, forms and instructions available	2	Forms, PDFs, instructions
Service links/forms functional	2	Non-destructive click-through tests
Contact or feedback channel available	1	Address, phone, email or contact form
Useful multilingual content where applicable	2	Arabic, French, Tamazight and English notes
Publication and update dates available	1	Dated content sample
Content clarity and cross-page consistency	1	Manual sample review

7.7 Technical Quality and Findability - 10 Points

Check	Points	Evidence
No significant broken internal links	2	Crawler and manual validation
No broken images/resources	1	Browser console/network evidence
Meaningful page titles and metadata	1	HTML head extraction
Sitemap/crawlability configuration	1	robots.txt, sitemap.xml
Clean URLs, canonicalization and redirects	1	URL and redirect samples
Functional site search and discoverability	2	Search task and indexability evidence
No critically outdated public technology	1	Public technology fingerprinting

Check	Points	Evidence
Cross-page technical consistency	1	Sample-page comparison

Each observable defect is assigned to one rubric row. Evidence may support contextual discussion elsewhere, but it cannot generate a second deduction unless a separately defined and independently observed user outcome is demonstrated.

8 Data-Protection Indicators

Data-protection indicators are reported separately from the technical score. They are evidence flags, not a legal compliance score. An external website review cannot establish full legal compliance. The review references Algerian Law No. 18-07 of 10 June 2018 on the protection of natural persons in personal-data processing, as amended and supplemented by Law No. 25-11 of 24 July 2025, together with current guidance from the National Authority for the Protection of Personal Data (ANPDP). Any legal conclusion requires specialist legal review before publication.

Status	Meaning
Observed	The defined public-facing indicator is present in the collected evidence.
Partially Observed	Some relevant public evidence exists, but defined elements are missing.
Not Observed	The indicator was not found in the documented sample; this is not proof of legal non-compliance.
N/M - Not Measured	There is not enough evidence for a reliable judgement.

The public-facing indicators checked where applicable are:

- privacy or personal-data notice availability;
- clear explanation of personal-data collection in forms or services;
- cookie, analytics, and third-party tracking transparency;
- contact channel for privacy-related requests;
- secure submission of personal data through HTTPS forms;
- consistency of privacy information across language versions.

9 Evidence, Reproducibility and Quality Assurance

9.1 Supplementary Technical Metadata

Supplementary metadata is collected to support analysis and interpretation, but it does not automatically add or remove points unless a rubric item explicitly depends on it:

- domain, canonical host, redirects, IP address, IPv4 and IPv6 support;
- ASN, hosting provider, hosting country, CDN and DNS provider;
- web server, CMS, framework, programming language where detectable;
- analytics systems, third-party services, cookies, TLS and certificate metadata.

9.2 Evidence Record

Every evaluated ministry should have an evidence folder or database record containing:

- evaluation timestamp, evaluator, corpus version and domain-verification evidence;
- tested URLs, page type labels, selected journeys and sampling notes;
- tool names, versions, settings and raw outputs;
- raw performance, accessibility, HTTP header and TLS results;
- uptime monitoring records;
- screenshots for desktop and mobile;
- manual notes for keyboard navigation, forms, citizen tasks, and content review;
- normalized result, points awarded, coverage status and score calculation sheet or JSON record;
- artifact paths or hashes, review status, conflicts, corrections and retest log.

Evaluator judgement items should be independently reviewed by a second evaluator. Disagreements are resolved using the written rubric and logged. A pilot on at least three diverse ministry sites is required before freezing the rubric; unclear items are revised before the national run, and any later scoring change requires a versioned update.

10 Safe Testing and Responsible Disclosure

The evaluation is limited to passive, non-destructive testing. The following are prohibited unless explicit written authorization is obtained:

- exploitation, authentication bypass, or SQL injection exploitation;
- password attacks, brute force, aggressive fuzzing, or denial-of-service testing;
- destructive testing or unauthorized vulnerability exploitation.

When sensitive exposure is found, the public report should describe the risk class and remediation direction without publishing exploit-ready details.

The team should maintain a private notification record and provide the affected institution a reasonable remediation window before publishing a sensitive finding. Public evidence packages must redact personal data, session identifiers, tokens, internal addresses, and unnecessary infrastructure detail.

11 Reporting and Deliverables

After the measurement phase, the study will produce a national ranking with score, coverage, status, observation date and stability information; a profile for every ministry; comparative category statistics; a machine-readable dataset; and a dashboard suitable for public analysis. Each ministry profile includes domain-verification evidence, tested pages and journeys, category results, separately reported data-protection indicators, supplementary metadata, major findings, evidence references and prioritized recommendations.

Comparative reporting includes category distributions and recurring conditions such as HSTS and CSP adoption, modern TLS support, Core Web Vitals, uptime, verified accessibility failures, broken-link patterns, technology use, hosting distribution, multilingual coverage and privacy-information availability. Dashboard filters distinguish Final, Provisional and Not Rankable results.

12 Standards and References

The methodology is aligned with ISO/IEC 25010:2023 as a product-quality reference model; WCAG 2.2 and WCAG-EM 2.0 for accessibility and representative sampling; OWASP and browser-security guidance for externally observable hardening; Core Web Vitals and web-performance guidance for performance; UN DESA digital-government benchmarking; and public-service guidance from GOV.UK and the U.S. General Services Administration. These sources inform the framework but do not make its 0–100 weighting an official standard.

- ISO/IEC 25010:2023 product quality model
- W3C Web Content Accessibility Guidelines 2.2
- W3C WCAG-EM 2.0
- OWASP Secure Headers Project
- MDN HTTP Observatory
- Lighthouse performance scoring and variability
- Google Core Web Vitals
- PageSpeed Insights documentation
- SSL Labs SSL Server Test
- Internet.nl
- UN E-Government Survey 2024
- GOV.UK Service Standard
- U.S. federal digital-services requirements checklist
- Algerian Law No. 18-07 of 10 June 2018
- Algerian Law No. 25-11 of 24 July 2025
- National Authority for the Protection of Personal Data
- Benaida 2023 Algeria/UK usability study
- ANPT summary of GAAN findings
- Official composition of the Algerian Government
- Ministry of Interior official-links directory
- Ministry of Foreign Affairs official-links directory

A Candidate Ministry Evaluation Corpus

The official Prime Minister composition page, checked on 3 September 2026, is the portfolio source of truth for this discovery pass. Older official link directories remain useful domain seeds but include legacy portfolio names and domains.

The operational seed list contains 34 ministry-portfolio entries: 32 candidate primary URLs and two unresolved entries. It is included here as a Phase 1 discovery appendix, not as the scored audit corpus. The Prime Minister, minister-wali, secretaries of state, delegated minister and Secretary General of Government are excluded unless the scope expands. After verification, this appendix

should be replaced or updated by the final evaluated corpus. The State-services inspectorate portfolio and the Ministry of Mines and Mining Industries require independent primary-domain confirmation.

Current portfolio	Candidate primary host	Discovery status
Foreign Affairs	mfa.gov.dz	Candidate
National Defence	mdn.dz	Candidate
Hydrocarbons	oilmines.gov.dz	Candidate
State-services inspectorate	—	Unresolved
Interior, local authorities and transport	interieur.gov.dz	Candidate
Justice	mjustice.gov.dz	Candidate
Finance	mf.gov.dz	Candidate
Higher education	mesrs.dz	Candidate
National education	education.gov.dz	Candidate
Health	sante.gov.dz	Candidate
Moudjahidine and rights holders	mmad.gov.dz	Candidate
Industry	industrie.gov.dz	Candidate
Pharmaceutical industry	miph.gov.dz	Candidate
Agriculture, rural development and fisheries	madr.gov.dz	Candidate
Mines and mining industries	—	Unresolved
Energy and renewable energies	energy.gov.dz	Candidate
Foreign trade and export promotion	mcepe.gov.dz	Candidate
Internal trade and market regulation	commerce.gov.dz	Candidate
Housing and spatial planning	mhuv.gov.dz	Candidate
Religious affairs and wakfs	marw.gov.dz	Candidate
Culture and arts	m-culture.gov.dz	Candidate
Youth	mjeunesse.gov.dz	Candidate
Post and telecommunications	mpt.gov.dz	Candidate
Knowledge economy and start-ups	startup.dz	Candidate
Communication	mcomm.gov.dz	Candidate
Vocational training	mvet.dz	Candidate
Public works and basic infrastructure	mtpib.dz	Candidate
Hydraulics	mh.gov.dz	Candidate
Labour, employment and social security	mtess.gov.dz	Candidate
Tourism and handicrafts	mta.gov.dz	Candidate
National solidarity and family	msnfcf.gov.dz	Candidate
Environment and quality of life	me.gov.dz	Candidate
Sports	msport.gov.dz	Candidate
Relations with Parliament	mrp.gov.dz	Candidate

Discovery reachability observations are not converted into scores. Browser verification, repeated checks and the complete monitoring protocol are required first.